Fear-Avoidance Beliefs Questionnaire

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PURPOSE

The FABQ focuses specifically on how a patient's fear-avoidance beliefs about physical activity and work may affect and contribute to his/her low back pain (i.e. the cognitive/affective components of pain that are differentiated from specific tissue damage, injury, and nociception) and resulting disability.

LINK TO INSTRUMENT

INSTRUMENT DETAILS

ACRONYM

FABQ

AREA OF ASSESSMENT

Activities & Participation
Bodily Functions
General Health & Development
Mental Functions
Sensation & Pain

ASSESSMENT TYPE

Patient Reported Outcomes

ADMINISTRATION MODE

Paper & Pencil

COST

Free

DIAGNOSIS/CONDITIONS

Pain Management, Spinal Cord Injury

POPULATIONS

Chronic Pain Mixed Conditions Musculoskeletal Conditions Spinal Cord Injuries

KEY DESCRIPTIONS

- Self-reported questionnaire consisting of 16 questions scaled from 0 to 6 (maximum score of 96; higher score indicates fear avoidance behaviors).
- The first 5 questions pertain to physical activity while the remaining 11 pertain to work.
- The Physical Activity subscale (FABQ-PA, range: 0 to 24) is the sum of items 2-5.
- The Work subscale (FABQ-W, range: 0 to 42) is the sum of items 6, 7, 9-12, and 15.

NUMBER OF ITEMS

16

EQUIPMENT REQUIRED

- Printed FABQ form

TIME TO ADMINISTER

5-10 minutes

DEPENDS ON THE PATIENT'S COGNITIVE ABILITIES

REQUIRED TRAINING

No Training

AGE RANGES

Adult	Elderly Adult
18 - 64	65 +
YEARS	YEARS

INSTRUMENT REVIEWERS

Initially reviewed by Amy Gwynn, Jayson Hull, Michael Irr, Nicholas Mang, Joseph Miller, Laura Rapp, Michelle Treffer, and Pat McNamara in September 2013; Updated by Natalie Mordini in June 2014

BODY STRUCTURE	ICF DOMAIN	MEASUREMENT DOMAIN
Back	Activity Participation	Activities of Daily Living Cognition General Health Sensory

PROFESSIONAL ASSOCIATION RECOMMENDATION

(George. Steven Z. 2006)

- Clinicians should emphasize methods of reducing fear-avoidance beliefs to effectively reduce pain intensity and disability for patients with acute low back pain.

Recommendations based on level of care in which the assessment is taken:

Recommendations for entry-level physical therapy education and use in research

CONSIDERATIONS

Pelvic Girdle Pain:

(Grotle et al, 2012)

- FABQ is not a good indicator of mental health for pelvic girdle patients, and it has questionable validity when assessing patients with pelvic girdle pain.

Patients with Chronic Low Back Pain:

(Calley et al, 2010, Chronic Low Back Pain)

- Therapists should not rely solely on their own judgments to identify self-reported fear-avoidance beliefs as it was found that questionnaires perform better.

(Crombez et al, 1999, patients with chronic low back pain)

- Pain-related fear measure turned out to be a better predictor of disability than pain intensity and the measure of general negative affect.
- FABQ-PA was significant and unique predictor of trunk extension/flexion peak torque.

(George, 2011)

- Do not give both the FABQ-PA and the TSK-11 due to the overlap in expected measures.
- Utilize FABQ-PA and PCS for LBP patients because both of these were loaded on separate factors in the factor analysis.
- Fear-avoidance beliefs about work was also loaded, so administer the FABQ-W if there are work concerns.
- FABQ-PA also consistently explained additional variance in pain intensity (18%) and disability (27%) after controlling for sex, age, and employment status.

(George. Steven Z. 2006)

- FABQ-PA responsiveness to clinically meaningful changes has not been validated.

(Cleland et al, 2007)

- FABQ-W might be appropriate to screen to identify patients with work-related LBP who are at risk for poor outcome with routine physical therapy.
- FABQ-PA nor FABQ-W were predictive of outcome for patients with private insurance.
- The use of FABQ as a screening tool for patients with non-work-related LBP was not supported.

Patients with Acute or Sub-Acute Low Back Pain:

(Beneciuk et al, 2012, Acute or Sub-acute Low Back Pain)

- Subgrouping based on multiple fear avoidance model measures may provide additional information on clinical outcomes in comparison with using the FABQ-PA alone.

Shoulder Pain:

(Mintkin et al, 2010)

- The FABQ (when "back" is replaced with "shoulder" in the measure) is an adequate predictor of how fear avoidance behaviors contribute to shoulder pain and disability. It is observed that the work pain beliefs subscale of items 8, 9, and 12-16 (specific to this study) had a + likelihood ratio >8 for higher scores (scores>12).

Workers with upper extremity injury:

(Inrig, 2012, Workers with UE injury)

- FABQ does not meet statistical standards for individual use as a screen

Musculoskeletal Disorders:

(Holden et al., 2010)

- "The clinical utility of FABQ-W scores for identifying participants at risk of non-return to work appears limited.
- The FABQ-W may have a clinical application in screening out patients unlikely to benefit from expensive and possibly unnecessary interventions.
- Individualised assessment and treatment may be the most effective way of preventing chronic non return to work."

Spine Surgery:

(Havakeshian & Mannion, 2013)

- "In a multivariable prospective (predictive) model, FABQ-PA was the only baseline psychological factor that significantly predicted outcome."

Cervical Radiculopathy:

(Dedering & Borjesson, 2012)

- The FABQ may be recommended for test-retest evaluations because "good" reliability was found, and it can discriminate between patients with cervical radiculopathy and healthy subjects.

Do you see an error or have a suggestion for this instrument summary? Please e-mail us!

Chronic Pain

STANDARD ERROR OF MEASUREMENT (SEM)

Pelvic Girdle Pain:

(Grotle et al, 2012)

- SEM = 2.20 for FABQ-PA

MINIMAL DETECTABLE CHANGE (MDC)

Pelvic Girdle Pain:

(Grotle et al, 2012 for "primary and secondary care clinics")

- MDC for individuals = 6.1 for FABQ-PA

- Group MDC = 6.1 for FABQ-PA

MINIMALLY CLINICALLY IMPORTANT DIFFERENCE (MCID)

Pelvic Girdle Pain:

(Grotle et al, 2012)

- 25% for FABQ-PA

CUT-OFF SCORES

Shoulder Pain:

(Mintkin et al, 2010)

Missing Work Prediction Metrics for Shoulder-Specific FABQ-WB (items 8,9, and 12-16)

FABQ-WB Score	Sensitivity	Specificity	+LR	-LR
0.5000	.75	.58	1.81	.42
1.5000	.75	.63	2.02	.40
2.5000	.62	.70	2.08	.55
3.5000	.62	.77	2.73	.49
5.0000	.62	.83	3.65	.45
6.5000	.50	.84	3.18	.59
7.5000	.50	.87	3.89	.57
8.5000	.50	.90	5.00	.56
9.5000	.50	.91	5.83	.55
10.5000	.37	.94	6.56	.66
12.5000	.37	.95	8.75	.65

14.5000	.37	.97	13.13	.64
16.5000	.25	.99	17.5	.76

NORMATIVE DATA

Neck Pain:

(Fau et al. 2006)

Table 1 Patient characteristics

Sample (centre)	A*(1,2,3)	B(1,3,4)	C(1,3)	D(1,2,3,4)
n	247	220	139	388
Patients (in full-time work)	171	168	107	337
Age				
Mean (years)	44.64	42.81	43.06	42.01
SD (years	9.83	9.99	9.99	9.51
Range (years)	18.0-68.0	18.0-68.0	19.0-68.0	18.0-68.0
	n (%)	*n* (%)	*n* (%)	*n* (%)
Male	91 (36.8)	80 (36.4)	58 (41.7)	150 (38.7)
Female	156 (63.2)	140 (63.6)	81 (58.3)	238 (61.3)
History				
1st episode	96 (38.9)	69 (31.4)	44 (31.7)	150 (38.7)
2nd or more episode	151 (61.1)	151 (68.6)	95 (68.3)	238 (61.3)
Pain duration before physiotherapy				

<1 month	88 (35.6)	97 (44.1)	58 (41.7)	151 (38.9)
1-3 months	50 (20.2)	47 (21.4)	27 (19.4)	93 (24.0)
3-6 months	40 (16.2)	32 (14.5)	19 (13.7)	51 (13.1)
>6 months	69 (27.9)	44 (20.0)	35 (25.2)	93 (24.0)
Pain patterns				
P1	47 (19.0)	45 (20.5)	26 (18.7)	79 (20.4)
P2	64 (25.9)	41 (18.6)	32 (23.0)	98 (25.3)
P3	136 (55.1)	134 (60.9)	81 (48.3)	211 (54.4)

*Patients recruited from centre 1 and 3 also participated in the study of construct validity. Centre 1: Prince of Wales Hospital. Centre 2: Alice Ho Miu Ling Nethersole Hospital. Centre 3: Queen Elizabeth Hospital. Centre 4: Quality Healthcare Asia Limited, Shatin Physiotherapy Clinic. P1, Neck pain only; P2, Neck pain + referred pain to upper limb(s); P3, Neck pain + neurogenic symptoms (pins and needles or numbness) ± referred pain to upper limb(s).

TEST/RETEST RELIABILITY

Pelvic Girdle Pain:

(Grotle et al, 2012)

- **Excellent** FABQ-PA (ICC = 0.88, 95% CI of 0.77-0.93)

Shoulder Pain:

(Mintkin et al, 2010)

- **Excellent** FABQ-PA (ICC = 0.88, 95% CI of 0.75-0.93).

Neck Pain:

(Fau et al. 2006)

- The mean time interval for the test-retest reliability was 12.5±7.8 days in sample A:

Table 2 Test-retest reliability and internal consistency coefficients for the fear-avoidance questionnaire in sample A patients

	Test-retest reliability ICC (95% CI)	*n*	Internal consistency (T) Cronbach's alpha (95% CI)
All	0.81 (0.75-0.86)	171	0.90 (0.88-0.92)
Age ≤ 40 years	0.72 (0.55-8.83)	64	0.83 (0.76-0.88)
Age > 40 years	0.85 (0.78-0.89)	103	0.92 (0.90-0.94)
Male	0.85 (0.75-0.90)	75	0.91 (0.88-0.94)
Female	0.78 (0.68-0.85)	96	0.89 (0.86-0.92)

Test-retest reliability: **Adequate-Excellent**

- Internal Consistency: **Excellent**

CRITERION VALIDITY (PREDICTIVE/CONCURRENT)

Predictive Validity:

Neck Pain:

(Fau et al. 2006)

- Cronbach's alpha for entire questionnaire = 0.90 (95% CI 0.88-0.92) (**Excellent**)

- The item total correlation coefficients for all 16 items of the questionnaire range: $r = 0.31$ to 0.68 (no item had to be excluded from further analysis) (**Adequate-Excellent**)

(George. Steven Z. 2006)

Correlations Among 4-week Changes in Disability, Fear-Avoidance Beliefs, and Average Pain Intensity:

- FABQ-PA and Disability: $r = .43$ (**Adequate**)
- FABQ-PA and Pain: $r = 0.45$ (**Adequate**)

CONSTRUCT VALIDITY

Shoulder Pain:

(Mintkin et al, 2010)

- Factor analysis identified a 4-factor solution (work pain beliefs for items 8, 9, and 12-16 [Cronbach α =0.89]
- Consequences of work for 6, 7, & 10-12 [α=0.88]
- Consequences of physical activity for items 2-5 [α=0.74], physical activity pain beliefs for items 1-2 [α=0.59]) with a cumulative response variance of 73.5% (explaining percentages of disability, as determined by SPADI). This forth factor was regarded as unstable.
- The FABQ-PA was identified as the strongest individual contributor to SPADI disability scores (when compared with these other scales and the TSK-11); it accounted for 11% of the variance in SPADI disability scores ($P<0.01$).

Pelvic Girdle Pain:

(Grotle et al, 2012)

The FABQ-PA had the following construct validity:

Test	Level of Validity	Spearman's r
Pelvic Girdle Questionnaire Total	Adequate	0.34
Pelvic Girdle Questionnare Activity Subscale (items 1-20)	Adequate	0.33
Pelvic Girdle Questionnaire Symptom Subscale (items 21-25)	Adequate	0.33
Disability Rating Index	Adequate	0.32

Oswestry Disability Index	Adequate	0.33
Pain Catastrophizing Scale	Poor	0.27
SF1 of SF-36 (General Health)	Poor	0.21
SF2 of SF-36 (Physical Functioning)	Poor	0.26
SF3 of SF-36 (Role-physical)	Poor	0.30
SF4 of SF-36 (Bodily Pain)	Poor	0.06
SF5 of SF-36 (Vitality)	Poor	0.25
SF6 of SF-36 (Social Functioning)	Poor	0.12
SF7 of SF-36 (Mental Health)	Poor	0.11
SF8 of SF-36 (Role-emotional)	Poor	0.13

- **Excellent** Discriminant validity for distinguishing pregnancy: AUC = 0.796 (95% CI of 0.701-0.890).
- **Adequate** Discriminant validity for distinguishing between pain at all 3 pelvic joints (2 SI and pubic symphysis): AUC = 0.592 (95% CI of 0.459-0.726).

(George. Steven Z. 2006)

Correlation of FABQ –PA Question and Average Pain Intensity During the Past 24 Hours

Question	**Level of Validity**	**Spearman's r**
PA makes my pain worse	Poor	0.25
PA might harm my back	Poor	0.21
I should not do PA that (might) make my pain worse	Poor	0.09
I cannot do PA that (might) make my pain worse	Poor	0.27

Neck Pain:

(Fau et al. 2006)

Table 3 Correlations between the fear-avoidance scores and other measures of pain, disability and health

Construct validity	rs	95% CI	*P*-value
Cross-sectional (sample B, *n* = 168)			
NRS			
At entry	0.34	0.20 to 0.47	<0.001
At discharge	0.33	0.20 to 0.46	<0.001
NPQ			
At entry	0.56	0.44 to 0.65	<0.001
At discharge	0.53	0.41 to 0.63	<0.001
SF-36 (physical)			
At entry	-0.45	-0.60 to 0.32	<0.001
At discharge	-0.64	-0.72 to 0.53	<0.001
SF-36 (mental)			
At entry	-0.36	-0.49 to 0.23	<0.001
At discharge	-0.43	-0.55 to 0.30	<0.001
Longitudinal (sample C, *n* = 107)			
Change in NRS			
At week 3	0.19	0.00 to 0.37	0.048

At week 6	0.18	-0.01 to 0.36	0.065
Change in NPQ			
At week 3	0.32	0.14 to 0.48	0.001
At week 6	0.38	0.20 to 0.53	<0.001
Change in SF-36 (physical)			
At week 3	-0.18	-0.36 to 0.01	0.065
At week 6	-0.27	-0.43 to 0.08	0.006
Change in SF-36 (mental)			
At week 3	-0.26	-0.43 to 0.08	0.006
At week 6	-0.24	-0.40 to 0.05	0.015

NRS, 11-point numerical rating scale; NPQ, Northwick Park Neck Pain Questionnaire; SF-36, Medical Outcomes 36-Item Short-Form Health Survey; rs, Spearman's correlation coefficient.

CONTENT VALIDITY

Neck Pain:

(Fau et al. 2006)

- All groups (A,B,C,D) 3.5-4.8 (**good-very good**)

FLOOR/CEILING EFFECTS

Pelvic Girdle Pain:

(Grotle et al, 2012): None for the FABQ-PA

Mixed Conditions

MINIMAL DETECTABLE CHANGE (MDC)

Workers with upper extremity injury:

(Inrig, 2012, Workers with UE injury)

- FABQ-W: 13 (change scores equivalent to 30-33% of scale)
- FABQ-PA: 8 (change scores equivalent to 30-33% of scale)

NORMATIVE DATA

Workers with upper extremity injury:

(Inrig, 2012, Workers with UE injury, *n* = 187, 56% working, 44% not currently working)

- FABQ-W: 35.2/42, SD = 6.7
- FABQ-PA: 20.3/24, SD = 4.4

TEST/RETEST RELIABILITY

(Waddell et al. 1993)

"All 16 individual items reached acceptable levels of test-retest reproducibility. 71% of individual answers were identical on retest which is high for 7-point scales. K statistics confirmed that all 16 items had high level of reproducibility. Two items had moderate concordance of 0.41-0.60, 8 had substantial concordance of 0.61-0.80 and 6 had close to complete concordance of greater than 0.80 (Landis and Koch 1977). The average level of K for all 16 items was 0.74 and all reached the 0.001 level of significance.

Workers with upper extremity injury:

(Inrig, 2012, Workers with UE injury)

- FABQ-W: **adequate** reliability (ICC = 0.55)
- FABQ-PA: **adequate** reliability (ICC = 0.69)

INTERRATER/INTRARATER RELIABILITY

(George. Steven Z. 2006)

- Interrater reliability: ICC = 0.94

INTERNAL CONSISTENCY

Workers with upper extremity injury:

(Inrig, 2012, Workers with UE injury)

- FABQ-W: **Good** consistency (Cronbach's alpha = 0.75)
- FABQ-PA: **Good** consistency (Cronbach's alpha = 0.78)

CONSTRUCT VALIDITY

(Waddell et al. 1993)

- FABQl was weakly related to sex.
- FABQ2 was not significantly related to sex.
- The mean score of FABQl in patients with low back pain alone was 21.9 +/- 14.3, in low back pain plus referred thigh pain 25.2 +/- 12.1 and in nerve root pain 17.8 +/- 12.0.
- Neither of the factors showed any significant correlation with total duration, duration of the present episode or time pattern of pain.
- Fear-avoidance beliefs about work were consistently more powerful in males than in females accounting for 27% and 35% of the variance of disability and work loss in the past year, respectively, in males but only 17% and 18% in females.

Pearson product-moment correlation coefficients for the 2 scales were:

- FABQ1: $r = 0.95$ (Excellent)
- FABQ2: $r = 0.88$ (Excellent).

Table III

Correlation of fear-avoidance beliefs and clinical variables

	Factor 1	Factor 2
Pain		
severity (VAS)	0.23 **	0.12
total duration	0.04	0.06
duration present episode	0.06	0.09
Disability		
activities of daily living	0.55 ***	0.51 ***
present work loss	0.39 ***	0.13

work loss in past year	0.55 ***	0.23 **
Psychological distress		
MSPQ	0.36***	0.19
depressive symptoms	0.41 ***	0.36 ***

** $P < 0.01$; *** $P < 0.001$; others not significant.

Workers with upper extremity injury:

(Inrig, 2012, Workers with UE injury)

Construct	Measure	Results	Correlation
Age	No correlation between age and FABQ subscales	FABQ-W rs = -0.05, P = 0.50 FABQ-PA rs = 0.01, P = 0.87	No Correlation
Pain intensity/ severity	SPADI pain subscale >0.4	FABQ-W rs = 0.24, No P = 0.0016 FABQ-PA rs = 0.23, P = 0.003	Poor correlation
	Von Korff pain intensity scale >0.4	FABQ-W rs = 0.25, P = 0.0015 FABQ-PA rs = 0.25, P = 0.001	Poor correlation
Physical function/ disability	QuickDASH >0.6	FABQ-W rs = 0.48, No P =<0.0001 FABQ-PA rs = 0.45, P =<0.0001	Adequate correlation

		FABQ-W rs = -0.18, P = 0.03	
	sf-MH >-0.4	FABQ-PA rs = -0.23, P = 0.003	Poor correlation
Mental health	sf-RE >-0.4	FABQ-W rs = -0.33, P =<0.0001	Adequate correlation
		FABQ-PA rs = -0.26, P = 0.001	Poor correlation
	MCS >-0.4	FABQ-W rs = -0.25, P = 0.0022	Poor correlation
		FABQ-PA rs = -0.30, P = 0.0002	
	# of days off work >0.4	FABQ-W rs = 0.31, No P =<0.0001	
		FABQ-PA rs = 0.17, P = 0.02	Adequate correlation
	Current work status	Wicoxon rank sum FABQ-W Z= 3.0497, P = 0.0027	Poor correlation
		FABQ-PA Z = 1.545, P = 0.1223	
Work related	WIS >0.40	FABQ-W rs = 0.46, P =<0.0001	Adequate correlation
		FABQ-PA rs = 0.38, P = 0.0002	
	DASH work module >0.40	FABQ-W rs = 0.51, P =<0.0001	Adequate correlation
		FABQ-PA rs = 0.42, P =<0.0001	

CONTENT VALIDITY

(Waddell et al. 1993)

- The present FABQ was developed in 2 pilot groups totaling 30 patients attending an orthopaedic out-patient clinic. The FABQ focused on patients' beliefs about how physical activity and work affected their current low back pain. It was based mainly on fear theory and fear-avoidance cognitions but also drew on the concept of Disease Conviction (Pilowsky and Spence 1975, 1983) which includes beliefs about the seriousness of the illness and its effect on the patient's life and on the concepts of somatic focusing and increased somatic awareness (Main 1983). The final format was a self-report questionnaire of 16 items presented on a single page (Appendix). The items were original though the wording of many of the items was derived from Fordyce's teaching aphorisms about pain behaviour (Fordyce 1988) and the 2 questions from Sandstrom and Esbjornsson (1986). Only I item was similar to 1 of the items in PAIRS (Riley et al. 1988) and 3 items to the SOPA harm scale (Jensen 1991). Each item was answered on a '7-point Likert scale from strongly disagree to strongly agree.

FACE VALIDITY

(Waddell et al. 1993)

- "There are several limitations to the present study. The analysis was based entirely on self-report measures and should ideally, as far as possible, be checked or compared with externally validated measures. These findings could be tested with alternative and more comprehensive measures of pain..." (pg.8)
- This paragraph sums up the measure of face validity. Not only is this measure entirely subjective, it is solely based on self report measures, therefore the face validity is in question.

(Cleland et al, 2007)

- The measure is based on self-report, so there are inherent face validity issues present.

FLOOR/CEILING EFFECTS

(Waddell et al. 1993)

- "In the complete series of 210 patients none had to be excluded because of lack of understanding or compliance on the FABQ itself, although 21 patients were unable to complete items 6-16 because they were not employed. Otherwise, there were only 1.3% missing answers and no individual item had greater than 2.1% missing answers. The answers to all 16 individual items were distributed across at least 4 categories and no item had to be excluded from further analysis because of excessively skewed distribution (Maxwell 1971)." (p. 6)

Workers with upper extremity injury:

(Inrig, 2012, Workers with UE injury)

- FABQ-W: **poor** ceiling effects (22.9%)
- FABQ-PA: **poor** ceiling effects (38.3%)

Chronic Pain

MINIMALLY CLINICALLY IMPORTANT DIFFERENCE (MCID)

LBP:

(George. Steven Z. 2006)

- 13-point change

NORMATIVE DATA

Patients with Acute or Sub-Acute Low Back Pain:

(Beneciuk et al, 2012, Acute or Sub-acute Low Back Pain, *n* = 108 with chronic low back pain of varying intensities)

- FABQ-W: 13.3/42, SD = 11.1
- FABQ-PA: 14.9/24, SD = 5.5

Chronic LBP:

(Cleland et al, 2007)

- FABQ-PA Private insurance: 14.6 (5.9), worker's compensation: 14.4 (6.6), significance: 0.77,
- FABQ-W: Private insurance: 12.9 (10.1), worker's compensation: 23.3 (10.8), significance: <0.001

TEST/RETEST RELIABILITY

Chronic LBP:

(Swinkels-Meewisse et al. 2003)

- FABQ/pa (r = 0.64; P ≤ 0.01; **Excellent**)
- FABQ/w (r = 0.80; P ≤ 0.01; **Excellent**)

(Staerkle et al. 2004)

- Kappa for all 16 items: 0.54 (range 0.34-0.68, P <0.01 for each coefficient; **moderate-substantial** agreement)
- Average weighted kappa: 0.76 (range 0.48-0.89, P<0.01; **moderate-almost**

perfect agreement)

- FABQ1 (r = 0.91; **Excellent**)
- FABQ2 (r = 0.84; **Excellent**)

INTERNAL CONSISTENCY

Chronic Low Back Pain:

(Crombez et al, 1999, patients with chronic low back pain)

- FABQ-W: **excellent** consistency (Cronbach's alpha = 0.84-0.92)
- FABQ-PA: **poor** consistency (Cronbach's alpha = 0.52-0.57)

(Swinkels-Meewisse et al. 2003)

- Cronbach Alpha: FABQ/pa for t1 and t2 were at1 = 0.70 and at2 = 0.72 (**Good**)
- Cronbach Alpha: FABQ/w were at1 = 0.82, and at2 = 0.83 (**Good**)

(Staerkle et al. 2004)

- Cronbach Alpha: FABQ1 (items 6, 7, 9, 10, 11, 12 and 15): 0.89 (**Excellent**)
- Cronbach Alpha: FABQ2 (items 2, 3, 4 and 5): 0.82 (**Excellent**)
- ICC: FABQ1: 0.91 (**Excellent**)
- ICC: FABQ2: 0.83 (**Excellent**)

CRITERION VALIDITY (PREDICTIVE/CONCURRENT)

Predictive Validity:

Chronic Low Back Pain:

(Crombez et al, 1999, patients with chronic low back pain)

- FABQ-W to pain disability (RDQ): **adequate** predictive value (standardized beta: 0.57)
- FABQ-PA to pain disability (RDQ): **adequate** predictive value (standardized beta: 0.40)
- FABQ-PA to trunk extension/flexion (TEF) torque: **poor** predictive value (standardized beta: -0.28)

LBP:

(George, 2011)

- Pain Intensity: Of assessed pain-related fear measures (TSK-11, PCS, FABQ-PA), only the FABQ-PA contributed statistically to the model, providing an additional 18% variance in pain intensity ratings ($P<0.01$). For disability scores, which the FABQ-PA provided an additional 27% variance ($p<0.01$).

(Swinkels-Meewisse et al. 2003)

Pearson's rho (concurrent):

- Tampa Scale for Kinesiophobia and FABQ/w for t1 $r = 0.33$ (**Adequate**) and t2 $r = 0.38$ (**Adequate**) ($P \leq 0.01$).
- TSK and FABQ/pa for t1 $r = 0.39$ (**Adequate**) and t2 $r = 0.59$ (**Adequate**) ($P \leq 0.01$)

(Cleland et al, 2007)

Table 8 Accuracy statistics for predicting a poor outcome using previously reported cut-off scores for the FABQ-PA and FABQ-W

	Sensitivity (95% CI)	Specificity (95% CI)	Positive likelihood ratio (95% CI)	Negative likelihood ratio (95% CI)
ABQ-PA: (cut-off > 13 points)				
Private insurance	0.60 (0.47, 0.71)	0.33 (0.26, 0.41)	0.89 (0.70, 1.13)	1.23 (0.83, 1.83)
Workers' compensation	0.70 (0.48, 0.86)	0.48 (0.35, 0.61)	1.46 (0.97, 2.18)	0.58 (0.28,1.19)
FABQ-W: (cut-off > 29 points)				
Private insurance	0.070 (0.028, 0.17)	0.92 (0.86, 0.95)	0.86 (0.28, 2.57)	1.01 (0.93, 1.11)
Workers' compensation	0.75 (0.53, 0.89)	0.83 (0.70, 0.91)	4.33 (2.27, 8.27)	0.30 (0.14, 0.65)

CONSTRUCT VALIDITY

LBP:

(George, 2011)

- FABQ-PA has **adequate** correlations with the FABQ-W (Pearson $r = 0.41$), TSK-11 ($r = 0.55$), PCS ($r = 0.31$), numeric pain rating scale ($r = 0.37$), and ODI ($r = 0.49$).
- FABQ-W has **adequate** corrections with the PCS (Pearson $r = 0.39$), numeric pain rating scale ($r = 0.37$), and ODI ($r = 0.34$); FABQ-W has a **poor** correlation with the TSK-11 ($r = 0.28$).

- Factor analysis identified items 2, 6, 7, 9, and 10-12 loaded on a pain-related fear of work (explained 12.26% of the variance) and items 2-5 loaded on a pain-related fear of physical activity (12.07% of variance), similar to the pain and work subscales

(Cleland et al, 2007)

- Private insurance: 0.20
- Worker's comp: 0.59

Chronic Low Back Pain:

(Crombez et al, 1999, patients with chronic low back pain)

Variables	Measures	Results	Correlation
Pain Intensity	VAS	FABQ-PA: r = 0.03	Poor
		FABQ-W: r = 0.41	Adequate
Pain Related Fear Measures	TSK	FABQ-PA: r = 0.57-0.76	Excellent to adequate
		FABQ-W: r = 0.53-0.56	Adequate
Negative Affect	NEM	FABQ-PA: r = 0.21-0.42	Poor to adequate
		FABQ-W: r = 0.35-0.38	Adequate
Disability	RDQ	FABQ-PA: r = 0.21-0.42	Adequate
		FABQ-W: r = 0.35-0.38	Excellent
Baseline Pain	GRS	FABQ-PA: r = 0.31	Adequate
		FABQ-W: r = 0.56	Excellent
Expected Pain Increase	GRS	FABQ-PA: r = 0.11	Poor
		PABQ-W: r = -0.04	Poor
Experienced Pain Increase	GRS	FABQ-PA: r = 0.18	Poor
		FABQ-W: r = 0.42	Adequate
Peak Torque	Trunk Ext/ Flex	FABQ-PA: r = -0.45	Adequate
		FABQ-W: r = -0.10	Poor

FLOOR/CEILING EFFECTS

LBP:

(Cleland et al, 2007)

- 189 patients were excluded from the study due to missing FABQ scores—it is possible that the missing scores may have influenced the results. Also, the duration of physical therapy was only 30 days, so it is difficult to project usefulness of the FABQ in terms of long-term prognosis.

RESPONSIVENESS

LBP:

(Staerkle et al. 2004)

- **Low to moderate**: FABQ1 (work, effect size: 0.33 [**Moderate** Change])
- FABQ2 (physical activity, effect size: 0.41 [**Moderate** Change])

Musculoskeletal Conditions

CUT-OFF SCORES

Musculoskeletal Conditions:

(Holden et al., 2010)

- FABQ-W cut-off score of > 39.5 and <= 27.5

Diagnostic properties for a FABQ-W cut-off score of >39.5 and <= 27.5

Diagnostic property	Score of > 39.5	Score <= 27.5
Sensitivity	31.4%	100%
Specificity	81.8%	16.7%
Positive predictive value	47.8%	61%
Negative predictive value	69.2%	100%
Sample prevalence	34.6%	34.6%

Positive likelihood ratio	1.7	1.2
Negative likelihood ratio	0.37	0

NORMATIVE DATA

Musculoskeletal Conditions:

(Holden et al, 2010, *n* = 117)

- FABQ-W score: 8–42, SD: 34.8 (6.5)

CRITERION VALIDITY (PREDICTIVE/CONCURRENT)

Predictive Validity:

Musculoskeletal Conditions:

(Holden et al, 2010)

- Study looked to "determine the predictive validity of the FABQ-W by establishing an upper FABQ-W cut off score with maximum specificity to identify participants at a high risk of nonreturn to work and a lower FABQ-W cut off score with maximumsensitivity to identify participants at low risk."
- Cutoff score 39.5: specificity=81.8%, positive predictive value=47.8%, & positive LR=1.7.
- Cutoff FABQ-W score >/=27.5: 100% sensitivity.

Spinal Cord Injuries

NORMATIVE DATA

Cervical Radiculopathy:

(Dedering & Borjesson, 2012)

	Patients: n=41	Healthy: n=41
FABQ-PA:	18 (4-24)	6 (0-21)
FABQ-W:	25 (0-42)	2 (0-27)
FABQ:	42 (12-66)	8 (0-41)

TEST/RETEST RELIABILITY

Cervical Radiculopathy:

(Dedering & Borjesson, 2012, *n* = 82 (k-value of weighted kappa and p-value of sign test))

- FABQ-PA: k=0.50 **Moderate**, p=0.243
- FABQ-W: k=0.67 **Good**, p=0.100
- FABQ: k=0.68 **Good**, p=0.617

INTERNAL CONSISTENCY

Cervical Radiculopathy:

(Dedering & Borjesson, 2012)

- FABQ: Cronbach's alpha 0.89, **Excellent**

CRITERION VALIDITY (PREDICTIVE/CONCURRENT)

Predictive Validity:

Spine Surgery:

(Havakeshian & Minnon, 2013)

- "In a multivariable prospective (predictive) model, FABQ-PA was the only baseline psychological factor that significantly predicted outcome."
- In lower FABQ-PA scores OR 0.877 (95 %CI 0.809–0.949), p = 0.001.

Cervical Radiculopathy:

(Dedering & Borjesson, 2012)

- "The present results indicate that the FABQ has concurrent validity also for patients with cervical radiculopathy because it is able to distinguish between patients and healthy subjects"
- (Mann-Whitney p-values: FABQ-PA, FABQ-W, and FABQ p<0.0001).

CONSTRUCT VALIDITY

Cervical Radiculopathy:

(Dedering & Borjesson, 2012)

The FABQ-PA, FABQ-W, and FABQ-SUM had the following construct validity:

Table 5. Spearman's correlation coefficients for the FABQ-SUM scores (FABQ-PA, FABQ-W, FABQ-SUM), Tampa scale sum score (TSK) and the scores of the questionnaires NDI, PCS, EQ-5D, GSES, HAD, physical activity and pain rated on a Borg CR-10 scale (n=41)

	FABQ-PA	FABQ-W	FABQ-SUM	TSK
TSK	*0.62***	0.32*	0.47**	
Borg CR-10	0.11	-0.17	-0.10	0.04
NDI	0.38*	0.45**	*0.51***	0.29
PCS				
Total	0.31*	0.07	0.17	*0.53***
Rumination	0.33*	-0.09	0.03	0.41**
Magnification	0.26	0.09	0.19	*0.52***
Helplessness	0.21	0.18	0.24	0.42**
EQ-5D				
Mobility	0.02	0.08	0.08	0.13
Self-care	0.29	0.35*	0.35*	0.17
Activity	0.26	0.41**	0.42**	0.14
Pain	0.47**	0.12	0.21	0.31*
Anxiety	0.19	0.19	0.23	0.32*
GSES	-0.03	-0.35*	-0.34*	-0.26
HAD				
Anxiety	0.19	0.17	0.22	0.39*
Depression	0.19	0.40**	0.42**	0.48**
Physical activity	-0.19	-0.43**	-0.45**	-0.22

*,** indicate correlation coefficients. Correlation coefficients exceeding 0.50 are in italics. FABQ, Fear Avoidance Beliefs Questionnaire; TSK, Tampa Scale for Kinesiophobia; NDI, Neck Disability Index; PCS, Pain Catastrophizing Scale; GSES, General-Self-efficacy Scale; HAD, Hospital Anxiety

RESPONSIVENESS

Cervical Radiculopathy:

(Dedering & Borjesson, 2012)

- "The present study focused on construct validity, but to determine further usefulness of the FABQ and the TSK questionnaires for patients with cervical radiculopathy, further studies addressing responsiveness are needed."

(George. Steven Z. 2006)

- Effect size = 1.8

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